Unit-5 : Cyber Security

5.1 Introduction to Cyber Security, Cyber Threats, Types of Cyber Attacks, Vulnerabilities, Intruders and Hackers, Threats: Worms, Virus, Ad- ware, Spy-ware, Trojans and covert channels, Backdoors, Bots, IP Spoofing, ARP spoofing, Session Hijacking

❖ Introduction to Cyber Security:

Cyber security is the practice of protecting systems, networks, and programs from digital attacks. These cyber attacks are usually aimed at accessing, changing, or destroying sensitive information.

The three main goals of cyber security:

- Confidentiality, Integrity, and Availability
- They work in harmony to protect data from threats, ensure its accuracy, and make it accessible to those who need it.

Types of Cyber security:

1. Network Security –

- It focuses on securing computer networks from unauthorized access, data breaches, and other network-based threats.
- It involves technologies such as Firewalls, Intrusion detection systems (IDS), Virtual private networks (VPNs).

2. Application Security:

- It is concerned with securing software applications and preventing vulnerabilities. It involves secure coding practices, regular software updates and patches, and application-level firewalls.

3. Information or Data Security:

- It focuses on protecting sensitive information from unauthorized access, disclosure, alteration. It includes Encryption, Access controls, Data classification, and Data loss prevention (DLP) measures.

4. Cloud Security:

- It involves securing data, applications, and infrastructure hosted on cloud platforms.It uses various cloud service providers such as AWS, Azure, Google Cloud, etc., to ensure security against multiple threats.

5. Mobile Security:

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- It involves securing the organizational and personal data stored on mobile devices such as cell phones, tablets, and other similar devices against various malicious threats. These threats are Unauthorized access, Device loss or Theft, Malware, etc.
- Regularly backing up mobile device data is important to prevent data loss in case of theft, damage, or device failure.
- Mobile devices often connect to various networks, including public Wi-Fi, which can pose security risks. It is important to use secure networks whenever possible, such as encrypted Wi-Fi networks or cellular data connections.

6. Endpoint Security:

- It refers to securing individual devices such as computers, laptops, smartphones, and IoT devices.
- It includes antivirus software, intrusion prevention systems (IPS), device encryption, and regular software updates.
- Antivirus and Anti-malware software that scans and detects malicious software, such as Viruses, Worms, Trojans, and Ransomware. Firewalls are essential components of endpoint security.

7. Internet of Things (IoT) Security:

- IoT security is the practice of protecting Internet of Things (IoT) devices from attack.
- It is based on a cybersecurity strategy to protect IoT devices and the vulnerable networks they connect to from cyber attacks.

❖ Cyber Threats:

A cyber threat or cyber security threat is a malicious act that seeks to damage data, steal data, or disrupt digital life in general.

Common types of cyber threats:

1. Malware
2. Man-in-the-middle attack
3. Denial-of-Service (DoS) Attacks
4. Phishing
5. Spoofing
6. Identity-Based Attacks
7. SQL Injection Attacks

Types of Cyber Attacks:

1. Malware Attacks:

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- Malware or malicious software is any program or code that is created with the intent to do harm to a computer, network or server.
- Malware is the most common type of cyber attack, It includes viruses, worms, trojans, spyware and ransomware.

Virus:

- Virus is a computer program or software that connects itself to another software or computer program to harm computer system.
- When the computer program runs attached with virus it perform some action such as deleting a file from the computer system. Virus can't be controlled by remote.
- Resident and Non-resident viruses are two types of Virus.

Worms:

- Worms is also a computer program like virus but it does not modify the program.
- Worm is a standalone program (malware) that can replicate itself and spread.
- It **replicate itself** more and more to cause slow down the computer system.
- Worms can be controlled by remote.
- A worm **consumes too much system memory**. Due to the replicating nature of worms, worms consume more system memory.

Trojan Horse:

- Trojan Horse does not replicate itself like virus and worms. It is a hidden piece of code which steals the important information of user. For example, Trojan horse software observe the e-mail ID and password while entering in web browser for logging.
- Trojan Horses generally install on the system as legitimate and useful software that can give unauthorized access and control of the system to the hackers. Back orifice, Rootkit and Beast Trojan are some of the common Trojan horse which is more harmful.
- Trojan Horse can be controlled by remote.
- The main objective of Trojan horse is to steal the information.

Spyware:

- Spyware is a form of malware designed to collect your personal information.
- The main objective of the spyware is to monitor the activity of the system.
- Spyware can be detected and removed by the anti-spyware program.
- Spyware does not replicate itself.
- Bonzibuddy, Cydore and Downloadware are some examples of spyware.

2. Man-in-the-Middle (MitM) Attacks

- A Man-in-the-Middle (MitM) attack occurs when an unauthorized entity intercepts the communication between two end points, such as a user and an application.
- The attacker gains the ability to eavesdrop on the communication, steal sensitive data, and assume the identity of each participant in the communication.

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- ✓ **Email hijacking:** Attacker pretends to be a legitimate organization, like a bank, to trick users into revealing sensitive info or transferring money.
- ✓ **DNS spoofing:** Manipulating Domain Name Servers to redirect users to a fake website, allowing the attacker to steal credentials or manipulate traffic.
- ✓ **IP spoofing:** Deceptive use of a false IP address to impersonate a website and deceive users into interacting with it.
- ✓ **HTTPS spoofing Exploiting**: the trust in HTTPS to make a malicious website appear secure, using "HTTPS" in the URL to mislead users.

3. Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks

- DoS and DDoS attacks aim to disrupt the normal functioning of a computer system, network, or website by overwhelming it with a large volume of traffic or requests.
- In a DoS attack, the traffic comes from a single source, while in a DDoS attack, multiple compromised devices (botnets) are used to launch the attack.
- These attacks are called 'denial of service" because the targeted website can't help users trying to access it.
- In a DoS attack, the site gets flooded with fake requests, using up its resources and causing it to shut down.
- Unlike other cyber attacks that aim to gain access, DoS and DDoS attacks just want to disrupt the target's service.
- To prevent DoS attacks, some use firewalls to detect and discard fake requests, ensuring normal traffic flows without interruption.

- **DoS attack techniques include**

- ✓ **HTTP flood DDoS-** the attacker uses HTTP requests that appear legitimate to overwhelm an application or web server.
- ✓ **UDP flood DDoS**-a remote host is flooded with User Datagram Protocol (UDP) packets sent to random ports.
- ✓ **ICMP flood** -a barrage of ICMP Echo Request packets overwhelms the target, consuming both inbound and outgoing bandwidth.

4. Phishing Attack

- Phishing attacks involve sending fraudulent emails or messages that appear to be from reputable sources to trick individuals into revealing sensitive information such as login credentials, credit card numbers, or personal information.
- Phishing attacks often use social engineering tactics to manipulate recipients into taking actions that benefit the attacker.
- The attacker sends emails pretending to come from a trusted source. **Phishing often involves sending fraudulent emails** to as many users as possible, but can also be more targeted.
- Example:
  - ✓ In phishing attack the hacker creates a fake web site that looks exactly like a popular site such as the SBI bank or paypal.
  - ✓ The phishing part of the attack is that the hacker then sends an e-mail message trying to trick the user into clicking a link that leads to the fake site.

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- ✓ When the user attempts to log on with their account information, the hacker records the username and password and then tries that information on the real site.
- ✓ Phishing emails will always tell you to click a link that takes you to a site where your personal information is requested. Legal organization never requests this information of you via email.

Different types of phishing attack are following:

- ✓ **Email Phishing:** Cybercriminals send deceptive emails, often pretending to be from a trustworthy source, to trick individuals into providing personal information or clicking on malicious links.
- ✓ **Spear Phishing:** This type targets specific individuals or organizations, tailoring the phishing attempt with personalized information to increase the chances of success.
- ✓ **Smishing (SMS Phishing):** Phishing attacks conducted through text messages (SMS) on mobile devices, where users are tricked into clicking on links or providing sensitive information.
- ✓ **Vishing (Voice Phishing):** Attackers use voice communication, often through phone calls, to deceive individuals into revealing sensitive information or taking specific actions.
- ✓ **Clone Phishing:** Attackers create a replica of a legitimate email, often from a trusted source, but with altered content or malicious links to deceive recipients.
- ✓ **Whaling:** Similar to spear phishing, but specifically targeting high-profile individuals, such as executives or CEOs, to gain access to valuable corporate information.
- ✓ **Search Engine Phishing:** Attackers create fake websites that appear in search engine results, tricking users into providing information on these fraudulent sites.

5. SQL Injection

- SQL injection is a one type of web attack used by hackers to steal (take) data from organization.
- SQL injection is a technique where malicious users can inject SQL commands into an SQL statement, via web page input.
- Injected SQL commands can alter (change) SQL statement and compromise the security of a web application.
- Since SQL statements are text only, it is easy, with a little piece of computer code, to dynamically change SQL statements to provide the user with selected data.
- SQL injection can be used to perform the following types of attack:
  -Authentication Bypass
  -Information Disclosure
  -Compromise Data integrity

Example:

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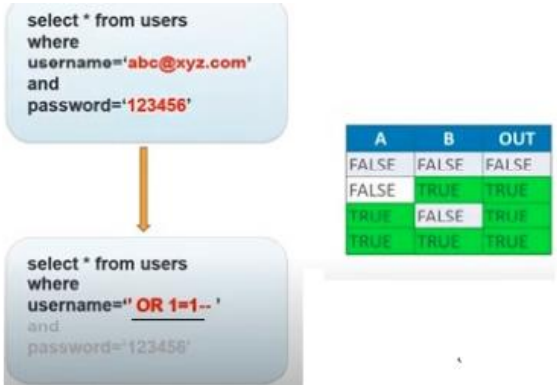

6. Supply Chain Attacks

- A supply chain attack is when cyber bad guys try to break into a system by targeting its connections. Instead of attacking a company directly, they go after its friends, like suppliers or partners.
- Imagine a store that gets products from different suppliers. Instead of breaking into the store directly, hackers might sneak into one of the suppliers and mess with the products there.
- Now, when the store gets those products, the problem comes along with them. So, the bad guys get into the store's system without attacking it directly. It's like getting into a club by pretending to be friends with someone already inside.

7. Cryptojacking

- Cryptojacking is when someone secretly uses your computer or device to mine crypto currency, like Bitcoin. It happens without your permission or knowledge.
- The attackers take advantage of your device's processing power to solve complex math problems, earning them crypto currency.
- You might notice your device slowing down or overheating. It often occurs through infected websites or malicious software.
- To protect against Cryptojacking, it's important to use strong antivirus programs and be cautious about the websites you visit. Regularly updating your software and browser also helps. Being aware of your device's performance can help you detect and stop Cryptojacking.

8. Zero-Day Exploits

- Zero-day exploits target previously unknown vulnerabilities (zero-day vulnerabilities) in software or hardware for which no patch or fix is available. Attackers exploit these vulnerabilities before they are discovered or patched by the vendor, making them particularly dangerous.
- It's called "zero-day because the defenders (the good guys) have zero days to fix or protect against the problem.
- It's like trying to find hidden treasure in a big field before someone else does.

9. Ransomware Attacks:

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Ransomware is a type of malware that encrypts files or locks down systems and demands payment (usually in cryptocurrency) from the victim in exchange for decryption keys or restoring access. Ransomware attacks can cripple organizations by encrypting critical data and disrupting operations.

10. Eavesdropping (packet sniffing)

- Eavesdropping like secretly listening to someone else's conversation without knowing them. In the digital world, it's when hackers intercept and listen to your private online chats or messages.
- For eg. If you're sending sensitive information like passwords over an unsecured Wi-Fi network, an eavesdropper could capture and misuse that data.
- For prevention, use encrypted connections, avoid public Wi-Fi.

11. Drive-By Downloads

- A drive-by attack is like catching a computer virus by just visiting a website, without clicking or downloading anything, Imagine you're strolling on the internet street, and a sneaky website infects your computer in the background.

- For example, a seemingly harmless site might secretly inject harmful code into your device, causing issues or stealing information. Drive-by attacks often exploit vulnerabilities in your browser or plugins.
- To protect yourself, keep your browser and software up to date, use reliable security tools, and avoid suspicious websites. It's like staying alert while wandering the digital neighborhood to avoid unexpected troubles.

12. Password Attack

- A password attack is when someone tries to figure out your password without your permission. It's like a secret code that helps you access your online accounts, and attackers want to crack it.
- Imagine if someone repeatedly tries different words or numbers to guess your password, like trying "123456" or "password." They might also use special programs to guess many combinations quickly.
- If they succeed, they could get into your accounts and do things you don't want. To stay safe, it's essential to use strong and unique passwords, avoid easily guessable choices, and change your passwords regularly.

13. DNS Spoofing

- DNS Spoofing, also known as DNS Cache Poisoning, is a cyber attack where an attacker manipulates the Domain Name System (DNS) to redirect traffic from a legitimate website to a malicious one.
- For example, you type "bank.com," but the hacker sends you to a fake bank site to steal your information. It's like a digital road sign tricking you into the wrong destination.
- To stay safe, use trusted networks, and be cautious if websites look different or ask for sensitive information unexpectedly.

❖ Vulnerabilities:

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Vulnerability in security refers to a weakness or opportunity in an information system that cybercriminals can exploit and gain unauthorized access to a computer system.
Vulnerabilities weaken systems and open the door to malicious attacks.

Hardware Vulnerability:
Hardware vulnerability is a weakness which can used to attack the system hardware through physically or remotely. For examples:
1. Old version of systems or devices
2. Unprotected storage
3. Unencrypted devices, etc.

Software Vulnerability:
A software error happen in development or configuration such as the execution of it can violate the security policy. For examples:
1. Lack of input validation
2. Unverified uploads
3. Cross-site scripting
4. Unencrypted data, etc.

Network Vulnerability:
A weakness happen in network which can be hardware or software. For examples:
1. Unprotected communication
2. Malware or malicious software (e.g.:Viruses, Worms, etc)
3. Social engineering attacks
4. Misconfigured firewalls

Human vulnerabilities:
Common human vulnerabilities includes;
1. Opening email attachments infected with malware
2. Forgetting to install software updates on mobile devices.

Intruders and Hackers:

- Intruder, often referred to as a hacker or cracker
- Intruders can be classified in three Classes:

Masquerader (Outsider):
- An individual who is not authorized to use the computer and who enter a system's access controls to use a legal user's account

Misfeasor (Insider):

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- A legal user who accesses data, programs, or resources for which such access is not authorized, or who is authorized for such access but misuses his or her privileges

Clandestine user(Insider or Outsider):

- An individual who take supervisory control of the system and uses this control to evade(avoid) auditing and access controls or to suppress audit collection
- (It exploiting (make use of) bugs in privileged programs.)

❖ Threats:

(Worms, Virus, Ad- ware, Spy-ware, Trojans and covert channels, Backdoors, Bots, IP Spoofing, ARP spoofing, Session Hijacking)

Virus:

- Virus is a computer program or software that connects itself to another software or computer program to harm computer system.
- When the computer program runs attached with virus it perform some action such as deleting a file from the computer system. Virus can't be controlled by remote.
- Resident and Non-resident viruses are two types of Virus.

Worms:

- Worms is also a computer program like virus but it does not modify the program.
- Worm is a standalone program (malware) that can replicate itself and spread.
- It **replicate itself** more and more to cause slow down the computer system.
- Worms can be controlled by remote.
- A worm consumes too much system memory. Due to the replicating nature of worms, **worms consume more system memory.**

Adware:

- Adware is software that displays unwanted advertisements on a user's device which are not always malicious; it can impact user experience and compromise privacy. **(**Adware is a type of malware that displays unwanted advertisements on your computer or device.)
- Adware works by installing itself quietly onto your devices – accidentally or otherwise – click on an advert that it displays to you.This is because, ultimately, adware exists to make money.
- Adware creators and distributing vendors make money from third parties.
- Adware can also track your search and browsing history to display ads that are more relevant to you.

Spyware:

- Spyware is a form of malware designed to collect your personal information.
- The main objective of the spyware is to monitor the activity of the system.
- Spyware can be detected and removed by the anti-spyware program.
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Backdoors:

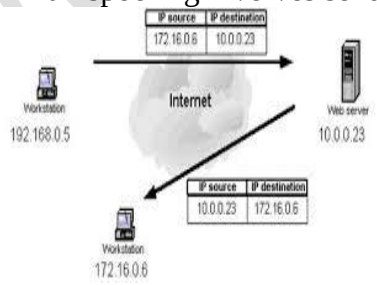
- Backdoors are hidden entry points into a computer system. It bypass normal authentication mechanisms.
- Attackers use backdoors to gain unauthorized access to systems.
- A Backdoor can be installed on your system by hackers in the form of some malware application or using your device's software vulnerabilities.

Bot:

- A bot, short for "robot", is a type of software application or script that performs automated tasks on command.
- Bots perform malicious tasks that allow an attacker to remotely take control over an affected computer. Once infected, these machines may also be referred to as zombies/Botnet.

IP Spoofing:

- IP spoofing is the creation of Internet Protocol (IP) packets which have a modified source address in order to either hide the identity of the sender, to impersonate another computer system, or both.
- In a spoof attack, the hacker modifies the source address of the packets which are sending, so that they appear to be coming from someone else.
- If the attacker obtains control over the software running on a network device, they can then easily modify the device's protocols to place an arbitrary IP address into the data packet's source address field. This is known as IP spoofing.
- Email spoofing involves sending message from a bogus email address.



ARP spoofing:

- ARP spoofing is a type of attack in which a malicious actor sends false ARP (Address Resolution Protocol) messages over a local area network.
- This results in the linking of an attacker's MAC address with the IP address of a legitimate computer or server on the network.

5.2 Cyber Crimes, Types of Cybercrime, Hacking, Attack vectors, Cyberspace and Criminal Behavior, Traditional Problems Associated with Computer Crime

❖ Cyber Crimes

Cybercrime is any criminal activity that involves a computer, network or networked device.

Types of Cybercrimes:

Identity Theft:

- Taking and using someone's personal information without permission. It's usually for doing fraud.
- For eg. Stealing someone's social security number to open bank account or get credit cards.

Hacking:

- Criminal hacking is the act of gaining unauthorized access to data in a computer or network.
- Hackers steal data ranging from personal information and corporate secrets to government intelligence. Hackers also break into networks to disrupt operations of companies and governments.

Ransomware:

- Ransomware attacks are a very common type of cybercrime.
- It is a type of malware that has the capability to prevent users from accessing all of their personal data on the system by encrypting them and then asking for a ransom to give access to the encrypted data. (It is a Malicious software locks valuable digital files and demanding a ransom for their release.)
- Ransomware typically affect via email, attract a user to click on an attachment or visit a website that infects their computer with malicious code.

Phishing and scams:

- Phishing refers to a method used by cyber criminals to obtain confidential information using emails or texts.
- Scammers pose as a trusted source (often a bank or well-known company) and trick recipients into providing personal information, such as account passwords and social security numbers. Phishing messages often use fraud link to attract recipients to click a link or open an attachment.

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Internet Fraud:

Internet fraud is a type of cybercrimes that makes use of the internet and it can be considered a general term that groups all of the crimes that happen over the internet like spam, banking frauds, theft of service, etc.

❖ Hacking:

Hacking is the act of gaining unauthorized access to data in a system or computer.

Hacking refers to activities that seek to compromise digital devices, such as computers, smartphones, tablets, and even entire networks.

Types of Hacking/hackers:

Positive hacking/Ethical hacking:

- It is also known as White Hat Hackers.
- White hat hackers are the one who is authorized or the certified hackers who work for the government and organizations by performing penetration testing and identifying loopholes in their cybersecurity.
- They also ensure the protection from the malicious cyber crimes.
- They work under the rules and regulations provided by the government, that's why they are called Ethical hackers or Cybersecurity experts.

Black Hat Hackers(Malicious hacking):

- They are often called Crackers.
- Black Hat Hackers can gain the unauthorized access of your system and destroy your vital data. The method of attacking they use common hacking practices they have learned earlier.
- They are considered to be as criminals and can be easily identified because of their malicious actions.

Gray Hat Hackers:

- Gray hat hackers fall somewhere in the category between white hat and black hat hackers.
- They are not legally authorized hackers.
- They work with both good and bad intentions; they can use their skills for personal gain. It all depends upon the hacker.
- If a gray hat hacker uses his skill for his personal gains, he/she is considered as black hat hackers.

Script Kiddies:

- They are the most dangerous people in terms of hackers.
- A Script kiddie is an unskilled person who uses scripts or downloads tools available for hacking provided by other hackers.
- They attempt to attack computer systems and networks and deface websites. Their main purpose is to impress their friends and society. Generally, Script Kiddies are juveniles who are unskilled about hacking.

Hacktivist/Hactivism:

- These are also called the online versions of the activists.

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- Hacktivist is a hacker or a group of anonymous hackers who gain unauthorized access to government's computer files and networks for further social or political ends.

❖ Attack vector:

- An attack vector is a pathway or method used by a hacker to illegally access a network or computer in an attempt to exploit system vulnerabilities.
- OR An attack vector is a path or means by which an attacker or hacker can gain access to a computer or network server in order to deliver a payload or malicious outcome.
- Attack vectors enable hackers to exploit system vulnerabilities, including the human element.

Common cyber attack vectors includes:

The most common attack vectors include the following:

Name	Description	Objective
Phishing	Deceptive emails, messages, or websites designed to trick individuals into revealing sensitive information or clicking on malicious links	Stealing credentials, spreading malware, or gaining unauthorized access
Malware	Malicious software, including viruses, worms, Trojans, and ransomware, designed to infect and compromise systems	Data theft, system damage, or unauthorized access
Social Engineering	Manipulating individuals to disclose confidential information or perform actions that compromise security.	Gaining unauthorized access, obtaining sensitive information, or facilitating other attacks.
DoS and DDoS	Overloading a system, network, or service with traffic to disrupt normal functioning	Rendering services unavailable, causing downtime, or overwhelming resources
Man-in-the- Middle (MitM)	Intercepting and altering communication between two parties without their knowledge.	Eavesdropping, stealing information, or injecting malicious content.
SQL Injection	Exploiting vulnerabilities in a database by injecting malicious SQL code	Unauthorized access, data manipulation, or extraction of sensitive information
Cross-Site Scripting (XSS)	Injecting malicious scripts into websites viewed by others	Stealing session cookies, redirecting users, or delivering malware
Zero-Day Exploits	Exploiting vulnerabilities in software or hardware that vendors are not aware of or have not released patches for	Gaining unauthorized access or compromising system integrity

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Password Attacks	Attempting to gain unauthorized access by guessing or cracking passwords	Unauthorized access to user accounts or systems
Brute Force Attacks	Repeatedly trying all possible combinations to guess a password or encryption key	Gaining unauthorized access through trial and error
Drive-By Downloads	Automatically downloading malicious software onto a user's device without their knowledge	Installing malware, stealing information, or gaining control over the system
DNS Spoofing	Manipulating the Domain Name System. (DNS) to redirect users to malicious websites	Phishing, spreading malware, or intercepting sensitive information
Bluetooth Hacking	Exploiting vulnerabilities in Bluetooth connections to gain unauthorized access to devices	Unauthorized access, data theft, or device manipulation
USB-Based Attacks	Using malicious USB devices to exploit vulnerabilities in systems when connected	Installing malware, stealing data, or gaining unauthorized access
Wi-Fi Eavesdropping	Monitoring and intercepting data transmitted over unsecured Wi-Fi networks	Eavesdropping, stealing sensitive information, or unauthorized access

❖ Cyberspace and Criminal Behaviour

- Cyberspace refers to the interconnected environment of computer systems, networks, and digital technologies where information is created, stored, processed, and exchanged.
- Here are some key aspects of cyberspace.
  - ✓ **Digital Realm:** Cyberspace exists in the digital realm, comprising electronic data and information accessible through computers and digital devices.
  - ✓ **Internet Infrastructure:** The internet serves as a significant component of cyberspace, connecting a vast network of computers and enabling global communication.
  - ✓ **Information Exchange:** Cyberspace facilitates the exchange of information through various online platforms, including websites, social media, emails, and messaging services.
  - ✓ **E-Commerce:** Online transactions, shopping, and financial activities are conducted within cyberspace through secure electronic payment systems.
  - ✓ **Cloud Computing:** Services such as cloud storage and computing contribute to cyberspace, allowing users to store and access data remotely.
  - ✓ **Virtual Communities:** Social networks, forums, and online communities provide spaces for people to interact and share information within cyberspace.
  - ✓ **Digital Innovation:** Cyberspace is a hub for technological innovation, fostering the development of new applications, software, and digital solutions.
  - ✓ **Internet of Things (IoT):** Connected devices, part of the lot, contribute to the expansion of cyberspace by creating an extensive network of interconnected smart devices.
  - ✓ **Cyber Laws and Governance:** Legal frameworks and regulations govern activities within cyberspace, addressing issues such as privacy, data protection, and cybercrimes.
  - ✓ **Cyber Threats and Risks:** As a dynamic environment, cyberspace is vulnerable to various threats, including cyber attacks, data breaches, and online fraud.

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- Criminals use the internet's secrecy, connection, and global reach to do various illegal activities. Here are some ways in which cyberspace intersects with criminal behavior.
  - ✓ **Cyber Theft and Fraud:** Cyber criminals engage in online theft and fraud by exploiting vulnerabilities in systems or using deceptive practices to trick individuals or organizations into providing sensitive information.
  - ✓ **Identity Theft:** Cyber criminals may steal personal information, such as social security numbers or bank details, to impersonate someone else for financial gain or to commit other crimes.
  - ✓ **Malware Attacks:** Malicious software, including viruses, worms, Trojans, and ransomware, is deployed to compromise computer systems, steal information, or demand a ransom.
  - ✓ **Phishing and Social Engineering:** Phishing attacks involve using deceptive emails, messages, or websites to trick individuals into revealing sensitive information, such as usernames, passwords, or credit card details.
  - ✓ **Cyber Espionage:** Nation-states or corporate entities may engage in cyber espionage to steal sensitive information, trade secrets, or intellectual property.
  - ✓ **Online Harassment and Cyber Bullying:** Criminals may use cyberspace to harass individuals, engage in cyber bullying, or spread false information, causing emotional distress or reputational harm.
  - ✓ Human Trafficking and Exploitation Cyberspace can be used for recruiting, advertising, and coordinating human trafficking activities, exploiting vulnerable individuals.
  - ✓ **Drug Trafficking and Dark Web Markets:** Illicit drug trade and other illegal activities often occur on the dark web, utilizing crypto currencies for transactions and maintaining a degree of anonymity.
  - ✓ **Ransomware Attacks:** Criminals use ransomware to encrypt a user's files, demanding payment (usually in crypto currency) for their release.
  - ✓ **Financial Cybercrimes:** Online banking fraud, credit card fraud, and other financial crimes exploit vulnerabilities in digital financial systems.
  - ✓ **Intellectual Property Theft:** Cyber criminals may steal and exploit intellectual property, including copyrighted material.
  - ✓ **Cyber terrorism:** Criminals or groups may use cyberspace for politically motivated attacks, aiming to disrupt critical infrastructure, create fear, or promote a particular agenda.

❖ Traditional Problems Associated with Computer Crime

- Traditional problems associated with computer crime, also known as cyber security challenges, have evolved alongside the rapid advancement of technology. Here are some longstanding issues

1. **Unauthorized Access and Hacking**
   - ✓ **Problem:** Unauthorized individuals gaining access to computer systems, networks, or databases
   - ✓ **Impact:** Breach of sensitive information, loss of data integrity, and potential disruption of services
2. **Malware Attacks**
   - ✓ **Problem:** Malicious software (viruses, worms, Trojans, ransomware) designed to infect and damage computer systems
   - ✓ **Impact:** Data loss, system damage, financial losses, and potential compromise of personal information
3. **Phishing and Social Engineering**
   - ✓ **Problem:** Deceptive tactics used to trick individuals into revealing sensitive information
   - ✓ **Impact:** Identity theft, financial losses, and unauthorized access to accounts

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4. **Identity Theft**
   - ✓ **Problem:** Unauthorized use of personal information to commit fraud or other criminal activities
   - ✓ **Impact:** Financial losses, damage to credit history, and potential legal issues for the victim
5. **Denial of Service (DoS) and Distributed Denial of Service (DDoS) Attacks**
   - ✓ **Problem:** Overwhelming a system, network, or website with traffic to disrupt normal functioning
   - ✓ **Impact:** Service downtime, financial losses, and potential damage to an organization's reputation
6. **Data Breaches**
   - ✓ **Problem:** Unauthorized access to and disclosure of sensitive information
   - ✓ **Impact:** Compromised customer data, loss of trust, and legal consequences
7. **Insider Threats**
   - ✓ **Problem:** Employees or trusted individuals intentionally or unintentionally causing harm to an organization's security
   - ✓ **Impact:** Unauthorized access, data breaches, and compromised intellectual property

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